

# Lecture 6 - P and PI Feedback

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## 6.1 P Control of 2nd Order System: Pendulum Example

Let's analyze at two operating points:  $\underbrace{\theta = 0}_{\text{down}}$  and  $\underbrace{\theta = \pi}_{\text{inverted}}$

1. Linearized model around  $\theta = 0$  (downward)

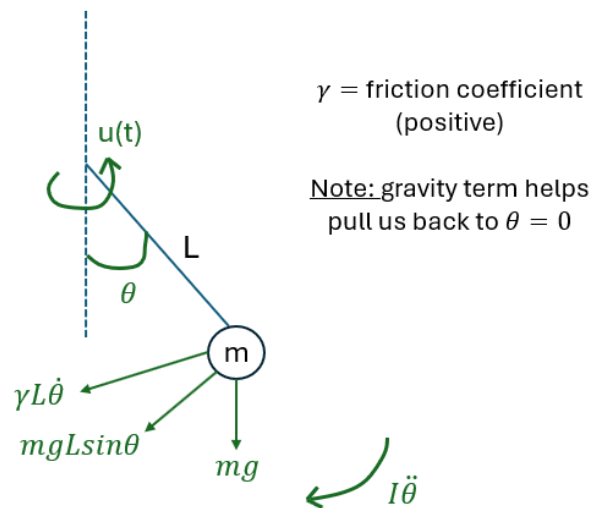


Figure 6.1: Forces on pendulum for  $\theta = 0$  state.

Equations of motion:

$$mL^2 \ddot{\theta} + \gamma L \dot{\theta} + mgL \sin \theta = u(t)$$

$$\ddot{\theta} + \frac{\gamma}{mL} \dot{\theta} + \frac{g}{L} \sin \theta = \frac{1}{mL^2} u(t)$$

Linearized model:

$$\ddot{\theta} + \frac{\gamma}{mL} \dot{\theta} + \frac{g}{L} \theta = \frac{1}{mL^2} u(t)$$

Transfer Function:

$$G(s) = \frac{1}{mL^2} \cdot \frac{1}{s^2 + \underbrace{\frac{\gamma}{mL}}_{a_1} s + \underbrace{\frac{g}{L}}_{a_2}}$$

Let  $m = L = 1$ ,  $g = 10$

$$\begin{aligned} \rightarrow G(s) &= \frac{1}{s^2 + \underbrace{\gamma}_{a_1}s + \underbrace{10}_{a_2}} \quad \left. \vphantom{\frac{1}{s^2 + \gamma s + 10}} \right\} \text{Stable since: } \begin{array}{l} \gamma > 0 \\ 10 > 0 \end{array} \\ \rightarrow \omega_n^2 &= 10 \quad \rightarrow \quad \omega_n = \sqrt{10} \\ 2\zeta\omega_n &= \gamma \quad \rightarrow \quad 2\zeta\sqrt{10} = \gamma \quad \rightarrow \quad \zeta = \frac{\gamma}{2\sqrt{10}} \\ \Rightarrow &\quad \begin{array}{l} \text{overdamped if } \zeta > 1 \text{ or } \gamma > 2\sqrt{10} \\ \text{underdamped if } \zeta < 1 \text{ or } \gamma < 2\sqrt{10} \end{array} \end{aligned}$$

2. Linearized model around  $\theta = \pi$  ( $y = \theta - \pi$ )

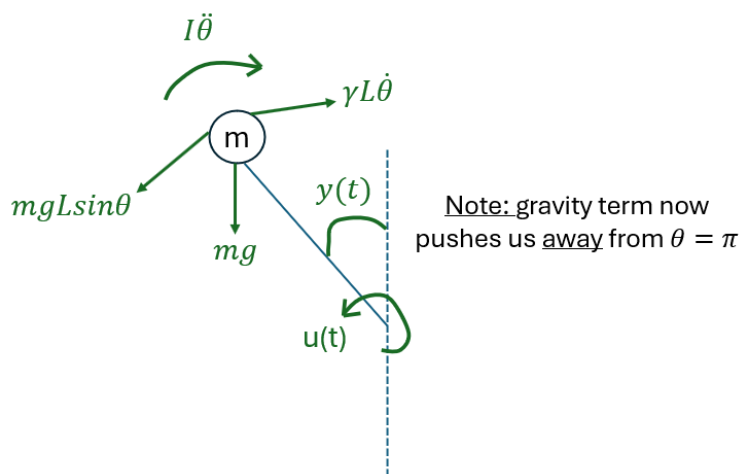


Figure 6.2: Forces on pendulum for  $\theta = \pi$  state.

Linearized equation:

$$\ddot{y} + \frac{\gamma}{mL}\dot{y} - \frac{g}{L}y = \frac{1}{mL^2}u(t)$$

Transfer function ( $m = L = 1$ ,  $g = 10$ ):

$$G(s) = \frac{1}{s^2 + \gamma s - 10}$$

$\rightarrow$  Unstable since  $-10 < 0$

### 6.1.1 Stabilize $\theta = \pi$ with P control

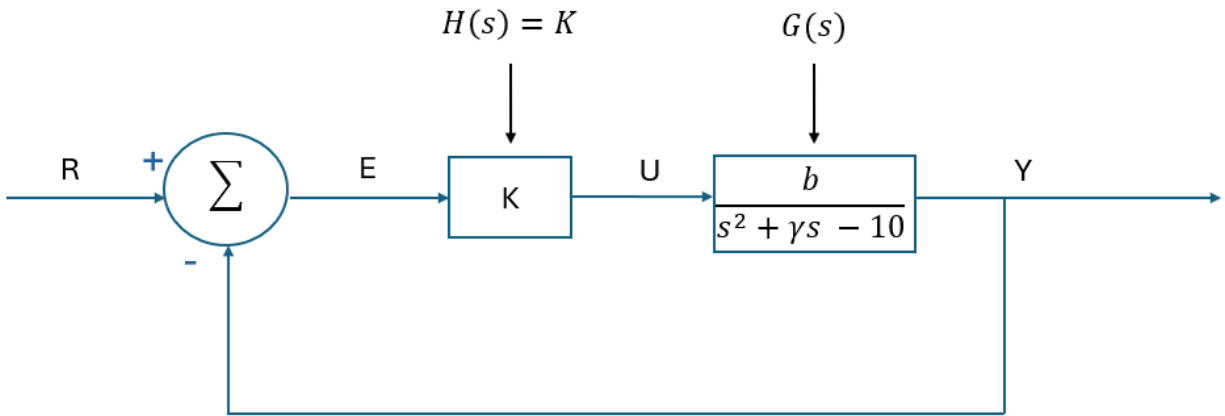


Figure 6.3: Block diagram for stabilizing the pendulum using P control.

$$G_{ry}(s) = \frac{GH}{1 + GH} = \frac{\frac{K}{s^2 + \gamma s - 10}}{1 + \frac{K}{s^2 + \gamma s - 10}} = \frac{K}{s^2 + \gamma s - 10 + K}$$

$$\implies \text{Stabilized if } K - 10 > 0$$

$$\rightarrow \text{Design } K > 10$$

Exercise: Choose  $K$  to set the natural frequency  $\omega_n = 5$

[Ans:  $\omega_n^2 = -10 + K = s^2 = 25 \rightarrow K = 35$ ]

How well does the P controller track a reference input?

Consider a small step input (small because our linearized model is only valid close to  $y = 0$ ):

$$r(t) = r \rightarrow R(s) = \frac{r}{s}$$

Final Value Theorem:

$$\begin{aligned} \lim_{t \rightarrow \infty} y(t) &= \lim_{s \rightarrow 0} sY(s) = s \underbrace{G_{RY}(s)R(s)}_{Y(s)} \\ &= s \cdot \frac{r}{s} \cdot \frac{K}{s^2 + \gamma s - 10 + K} \\ &= r \cdot \frac{K}{-10 + K} \end{aligned}$$

$\rightarrow$  For e.g.  $K = 35$ ,  $y(t) = r \cdot \frac{35}{25} = \frac{7}{5}r \neq r$

$\rightarrow$  Steady-state error  $E_{ss} > 0$  (no matter what  $K$  we chose)

Knowledge Check:

Which of the following transfer functions can produce this unit step response?

Hint:

- Check stability
- Check steady-state value  $G$
- Check shape of response

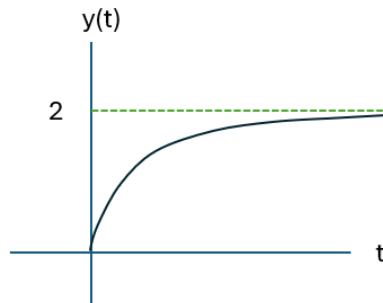


Figure 6.4: System response in the time domain.

1.  $\frac{2}{2+s}$   $\times$  FV = 1
2.  $\frac{1}{s^2+10s+1/2}$   $\checkmark$  FV = 2, real neg poles,  $b^2 - 4ac > 0$
3.  $\frac{1}{s^2+\frac{1}{10}s+1/2}$   $\times$  FV = 2 but oscillating ( $b^2 - 4ac < 0$ )
4.  $\frac{2}{1-s}$   $\times$  unstable;  $\frac{-2}{s-1}$

## 6.2 PI Controller

Let's revisit a simple cruise control example:

$$\dot{y}(t) + y(t) = u(t) + d(t)$$

Block diagram with feedback control:

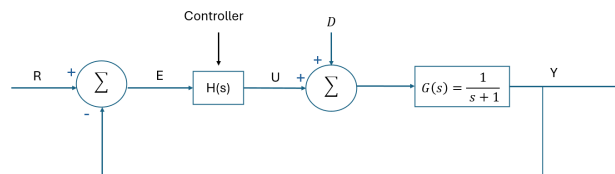


Figure 6.5: PI controller block diagram for cruise control problem.

Previously, we discussed P control (P = Proportional)

$$\begin{aligned} H(s) &= K_p && \leftarrow \text{Just a scalar.} \\ U(s) &= K_p \cdot E(s) && \leftarrow \text{The input is proportional to the error} \end{aligned}$$

Now we introduce PI control (Proportional + Integral)

$$\begin{aligned} H(s) &= K_p + \frac{K_i}{s} \\ U(s) &= E(s) \cdot (K_p + \frac{K_i}{s}) && \begin{array}{l} \leftarrow \cdot \text{ The error} \\ \quad \cdot \text{ PLUS The integral of the error} \end{array} \\ \downarrow \text{Inverse Laplace} \\ \mathcal{L}^{-1}(U(s)) = u(t) &= K_p e(t) + K_i \int_0^t e(t) dt \end{aligned}$$

→ Controller now also looks at how much error has accumulated over time.

→ This removes the bias error that P control had for constant inputs.

To see why bias is removed, let's look at the new transfer functions  $G_{ry}$  and  $G_{dy}$

$$\begin{aligned} G_{ry}(s) &= \frac{HG}{1+HG} = \frac{(K_p + \frac{K_i}{s})(\frac{1}{s+1})}{1 + (K_p + \frac{K_i}{s})(\frac{1}{s+1})} = \frac{\frac{sK_p + K_i}{s(s+1)}}{1 + \frac{sK_p + K_i}{s(s+1)}} \\ &\rightarrow \frac{sK_p + K_i}{s(s+1) + sK_p + K_i} = \boxed{\frac{K_p s + K_i}{s^2 + (K_p + 1)s + K_i}} \\ G_{dy}(s) &= \underbrace{\frac{G}{1+GH}}_{\text{same denom}} = \frac{\frac{1}{s+1}}{1 + \frac{sK_p + K_i}{s(s+1)}} = \boxed{\frac{s}{s^2 + (K_p + 1)s + K_i}} \end{aligned}$$

Let's Check Stability

→ Recall: for  $s^2 + a_1 s + a_2$  → stable iff  $a_1, a_2 > 0$ .

→ Stable iff:  $(K_p + 1) > 0$  and  $K_i > 0$

Let's Check Tracking and Regulation

SS response to constant inputs (reference  $r_0$  and disturbance  $d_0$ ) where

$$\begin{aligned} r(t) = r_0 &\xrightarrow[\mathcal{L}(\cdot)]{} R(s) = \frac{r_0}{s} \\ d(t) = d_0 &\xrightarrow[\mathcal{L}(\cdot)]{} D(s) = \frac{d_0}{s} \end{aligned}$$

FVT:

$$\begin{aligned}
 \lim_{t \rightarrow \infty} y(t) &= \lim_{s \rightarrow 0} sY(s) = s[G_{ry}(s)R(s) + G_{dy}(s)D(s)] \\
 &= s \cdot \frac{r_0}{s} \cdot \left( \frac{K_p s + K_I}{s^2 + (K_p + 1)s + K_I} \right) + s \cdot \frac{d_0}{s} \cdot \left( \frac{s}{s^2 + (K_p + 1)s + K_I} \right) \\
 &= r_0 \cdot \frac{K_I}{K_I} + 0 \\
 \boxed{y(t = \infty) = r_0} &\quad \leftarrow \quad \text{perfect tracking of constant reference, even with constant disturbance}
 \end{aligned}$$

### PI Controller Design

We can choose  $K_p$  &  $K_I$  to set the location of poles, i.e. to set  $\omega_n$  &  $\zeta$ . (subject to stability constants:  $K_p + 1 > 0$ ,  $K_I > 0$ )

→ Example

Want to set  $\omega_n = 2$ ,  $\zeta = 2$ . Together, these prescribe how fast/slow our modes are, and how fast our response decays.

$$\begin{aligned}
 \rightarrow K_I &= \omega_n^2 = 2^2 = 4 \\
 K_p + 1 &= 2\zeta\omega_n = 2 \cdot 2 \cdot 2 = 8 \quad \rightarrow \quad K_p = 7
 \end{aligned}$$